

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
CHANGA**

Second Semester of M.Tech. (CSE) Examination

May 2011

EC – 709 Digital VLSI Design

Date: 17.05.11, Tuesday

Time: 10.00 a.m to 1.00 p.m

Maximum Marks: 70

Instructions: (1) Attempt all questions.

(2) Answer to the two sections must be written in separate answer books.

(3) Figures to the right indicate full marks

(4) Make suitable assumptions and draw neat figures whenever required.

SECTION –I

Q-1 Answer the following questions. (any four) 12

- (a) Explain VLSI design flow in brief.
- (b) What is effect of channel length modulation on current-voltage characteristics of MOSFET? Explain with proper derivation.
- (c) Explain design issues of CMOS inverter.
- (f) Compare static power dissipation and switching power dissipation of CMOS inverter.
- (g) Why in transmission gate one pMOS and one nMOS are connected in parallel?
- (h) Explain oxide related capacitance in brief.
- (i) Explain Pseudo-nMOS Gates.

Q-2(a) What is gradual channel approximation? Derive the current equations for nMOS transistor. 6

- (b) Measured voltage and current data for MOSFET are given below. 4
Determine the type of the device and calculate the parameters k_n , V_{TO} , and γ . Assume Fermi potential is -0.3V

$V_{GS}(V)$	$V_{DS}(V)$	$V_{SB}(V)$	$I_D(\mu A)$
3	3	0	97
4	4	0	235
5	5	0	433
3	3	3	59
4	4	3	173
5	5	3	347

- (c) Define sub-threshold current. 1

OR

- Q-2(a)** Compare the two technology scaling methods, namely (i) constant electric field scaling and (ii) constant voltage scaling. In particular show analytically by using equations how the delay time, power dissipation and power density are affected in terms of the scaling factor S. 5
- (b)** Explain circuit operation of Depletion load nMOS inverter. Derive critical voltage points V_{IL} , V_{IH} , V_{OH} , V_{OL} for the same. 6

- Q-3(a)** Define the propagation delay τ_{PHL} and τ_{PLH} . Derive τ_{PLH} for CMOS inverter. 6
- (b)** Explain clocked SR latch with its gate level schematic and waveform. 6

OR

- Q-3(a)** Derive the threshold voltage expression for CMOS two input NOR Gate. 6
- (b)** Consider the following Boolean function 6

$$Z = \overline{A(D + E) + BC}$$

Realize Boolean function using nMOS complex logic gates and complex CMOS logic gates.

SECTION –II

- Q-4 (a)** Discuss basic steps for fabrication and briefly explain fabrication process of p-well CMOS. 5
- (b)** Discuss the impact of different VLSI design styles upon the design cycle time and the achievable circuit performances. 4
- (c)** Compare Gallium arsenide technology with CMOS technology. 2

- Q-5 (a)** Explain voltage Bootstrapping in detail 6
- (b)** Draw & Explain the equivalent circuits of DRAM and SRAM memory cells 6

OR

- Q-5 (a)** Explain Dynamic CMOS Logic (Precharge-Evaluate Logic) in detail. Also illustrate the cascading problem in dynamic CMOS logic. 6
- (b)** Draw Basic BiCMOS circuits and explain it in detail. 6

- Q-6** Write a short note on following (any three) 12
- (a)** BiCMOS applications.
- (b)** D-MESFET and E-MESFET
- (c)** Full custom Vs Semi- Custom design
- (d)** Short channel effect of MOSFET.
- (e)** Different isolation techniques used in Fabrication.